

GIHAN LAKMAL

AI / ML Engineer | Computer Science Specialized in AI | RAG, LLMs, MLOps & Multi-Agent Systems

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ABOUT

AI/ML Engineer with nearly 2 years of experience developing, deploying, and optimizing machine learning, deep learning, and generative AI solutions. Skilled in Python, data preprocessing, feature engineering, model evaluation, time-series forecasting, NLP, computer vision, RAG, LLM applications, and multi-agent AI systems. Experienced with FastAPI, LangChain, LangGraph, ChromaDB, Neo4j, PostgreSQL, Docker, MLOps, DevOps, CI/CD, monitoring, logging, model tracking, and performance optimization for scalable AI solutions and business-focused stakeholder insights across real-world production projects and deployments.

EDUCATION

University of Kelaniya - Sri Lanka (2022-2026)

B. Sc. (Hons) Computer Science (AI Specialization) (Final-Semester) - 3.61

University of Colombo - Sri Lanka (2022-2023)

Bachelor of Information Technology(External)

Nalanda College, Colombo 10

G.C.E. Advanced Level 2020 Physical Science Stream Combined Mathematics, Physics, Chemistry

Z: 1.563

PROJECTS

1. Medical Drug Sales Prediction based on Specific areas in Sri Lanka (IEEE Research Publication) [github](#)

Built production-grade time-series forecasting and inference pipeline for weekly, area-level pharmaceutical sales.

- Modeling: Architected and ensembled GRU/LSTM, Transformer/TFT, N-BEATS, LightGBM/XGBoost, and Prophet models with time-series-aware cross-validation.
- Feature Engineering: Implemented lag/rolling features, calendar/holiday encodings, seasonal decomposition, imputation, and scaling pipelines for robust sequence learning.
- Explainability & Causality: Delivered SHAP-based local/global explanations and causal discovery/counterfactual analyses to surface actionable drivers.
- Deployment & MLOps: Containerized inference/API (Docker, Flask/FastAPI), CI tests, experiment tracking, and monitoring for model drift and reproducibility.

2. FinTech MultiAgent System For StockBroker Firm - Vinvest Client Project [github](#)

Built Get-Deep — an AI platform that combines multi-agent LLM orchestration (LangChain/LangGraph), vector search (ChromaDB), and a Neo4j knowledge graph to provide retrieval-augmented conversational analytics and automated reporting.

- Architected and implemented a retrieval-augmented multi-agent system using FastAPI and LangChain/LangGraph to orchestrate specialized agents for general, deep-reasoning, graph-update, and report-generation workflows.
- Designed and built a scalable ingestion pipeline for heterogeneous documents (PDF/Word/Excel) with token-aware chunking, NER metadata enrichment, batched embeddings, and idempotent Neo4j upserts to avoid duplicates.
- Modeled and maintained a Neo4j knowledge graph (documents, entities, metrics) with indexed Cypher queries and transaction-safe updates to support entity-aware retrieval and reasoning.
- Implemented hybrid retrieval (Chroma vector search + Neo4j graph expansion) and prompt-assembly with tiktoken budgeting to improve answer relevance and reduce LLM API costs.
- Integrated observability and QA: LangSmith tracing for agent runs, structured logging with correlation IDs, health checks, and Docker Compose deployment for reproducible environments.
- Reduced operational cost and latency by batching embedding requests, adding async/background workers for heavy tasks, and caching frequent retrievals.

3. Advanced Agentic Hybrid OCR PROJECT - Client Project ArcheAI [github](#)

- Designed and deployed a production-grade OCR and AI-driven document processing system for structured data extraction from complex multi-page insurance approval documents. Built a 7-phase processing pipeline integrating Tesseract OCR, Vision-Language Models (VLM), automated validation, and AI-based gap filling to achieve 94-98% extraction accuracy. Implemented a feedback learning mechanism that learns from human corrections, reducing manual review rates from 40% to 10% over time without model retraining.
- Architected workflow orchestration using Temporal for reliable, scalable, multi-step processing with asynchronous workers. Developed REST APIs using FastAPI and designed a React-based review interface for human validation. Integrated PostgreSQL with optimized schema design for job tracking, extraction storage, and audit logging.
- Extended the system by routing structured outputs into a multi-agent orchestration framework combined with a Retrieval-Augmented Generation (RAG) pipeline, enabling intelligent user interaction, contextual querying, and dynamic response generation over processed documents. Deployed using Docker with production-ready configurations, multi-tenant support, security controls, and performance optimization achieving 100+ documents per hour throughput.
- Tech Stack:** Python, FastAPI, Temporal, PostgreSQL, OpenAI Vision Models, Tesseract OCR, React, Docker, SQLAlchemy.

EXPERINCE

ArcheAI - Co Founder (Dec 2025 - present) [Linkedln](#)

- Co-founded ArcheAI, an AI research and engineering startup focused on building production-grade intelligent systems combining machine learning, multi-agent orchestration, and scalable software infrastructure. Driving technical innovation by designing end-to-end AI solution architectures for enterprise and client-specific automation platforms. , Agentic Whatsapp Bot , Fax Extraction models , Agentic Story Telling Platforms.
- Led development of multiple client deployments from research prototyping to production rollout, delivering 3 full-cycle AI projects integrating document intelligence, workflow automation, and knowledge-driven reasoning systems. Architected modular backend and AI pipeline structures supporting extensible node-based computation and distributed task execution.
- Conducting applied research in foundation model adaptation, retrieval-augmented generation (RAG), semantic embedding optimization, and hybrid neuro-symbolic workflows to enhance contextual understanding and system reliability. Implementing DevOps and MLOps practices including containerized deployment, CI/CD integration, model serving strategies, and performance monitoring for high-availability AI services.

AI /ML Associate Engineer at SenzMate AIOT (july 2025 - present)

- Contributing to the development of the Interplay Autonomous Platform as a Node Engineer, designing and implementing modular AI nodes and workflow pipelines to enable scalable multi-agent automation for enterprise users. Building finance-focused multi-agent systems that coordinate task routing, memory management, and intelligent decision-making using advanced orchestration frameworks.
- Working on real-world client projects with end-to-end ownership, including system design, AI model integration, backend development, deployment, and production optimization. Leveraging LangChain, LangGraph, and modern AI orchestration frameworks to develop dynamic agent workflows and Retrieval-Augmented Generation (RAG) pipelines.
- Working with advanced AI/ML paradigms including foundation models, transformer-based architectures, contextual embedding spaces, semantic vector search, and probabilistic inference techniques, applying model optimization strategies, prompt engineering, and hybrid symbolic-neural workflow integration for intelligent automation systems.
- Independently managing full project lifecycles, from architecture to deployment, while applying DevOps and MLOps practices including containerization, environment configuration, performance monitoring, and scalable deployment strategies.
- **Tech Stack: Python, LangChain, LangGraph, Multi-Agent Systems, RAG, FastAPI, Docker, CI/CD, MLOps.**

AI Research Assistant Intern (November 2024 - June 2025)

Worked on AI projects focusing on building generative AI models for Sri Lankan native languages, including Sinhala and Tamil. Our team collected a corpus of 40 million Tamil and Sinhala words, contributing to language model development. I gained hands-on experience in data analysis and preprocessing, enhancing the models' accuracy and relevance for native speakers. The project involved generative AI, computer vision, and audio generation, applying real-life data for improved solutions.

SKILLS

Programing languages:

Python, Java, JavaScript, C, C#

Tools:

MySQL, PSQL , Jupyter Notebooks, Github, anaconda enviroment , VScode

Software Development:

OOP, Data Structures, Algorithms , Devops

Technologies:

SQL , NumPy, Pandas, TensorFlow, Keras, Scikit-learn, Seaborn, Python, Flask, ARIMA, SARIMAX , Gen AI, ML

Web Development:

HTML, CSS , JavaScript

AI & Data Science:

ML (Supervised/Unsupervised), NLP (Text Processing) , Gen AI ,Transformer Models , Prompt Engineering ,Multi-Agent Systems

Concepts:

Data Analysis, Data Visualization, Machine Learning, Deep Learning, Time-Series Forecasting, Predictive Modeling, Natural Language Processing (NLP), AI, Probability, Statistics, Data Preprocessing, Feature Engineering, Python Programming, ARIMA, SARIMAX , DVC , MLFlow , Docker , Grafana and prometheus , MLOps , CI/CD

REFERENCES

Arunn Balasundaram

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Prof. Nayoaml Dias

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